

Md Imam Ahasan

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🎓 Education

Chongqing University (CQU)

M.Sc. in Computer Science

GPA: 83.4/100 | Fully-funded by the Chongqing University President's Scholarship

Chongqing, China

Sep 2023 – Jun 2026

Daffodil International University (DIU)

B.Sc. in Computer Science and Engineering

GPA: 71.5/100 | Best Club Performance Award, Daffodil International University

Dhaka, Bangladesh

Sep 2017 – Sep 2021

📖 Thesis

Research on Low-Dose Computed Tomography Image Reconstruction

M.Sc., Chongqing University

Developed DACD and GenDiff, two physics-guided diffusion frameworks for LDCT reconstruction. Advisor: Dr. Guangchao Yang.

Gronthon: Bangla PDF to Text Conversion Application

B.Sc., Daffodil International University

Developed an end-to-end OCR pipeline for Bengali script digitization. Advisor: Dr. Md Zahid Hasan.

📚 Publications

Peer-Reviewed Conference/Journal Papers

- A F M Abdun Noor, **Md Imam Ahasan**, Mohammad Azam Khan, Guangchao Yang*. *GeGLUNet: Structural Retinal Vessel Segmentation via Attention-Gated GeGLU and Contrastive Supervision* - Conference on Pattern Recognition and Computer Vision (PRCV 2025) - **CCF-C**. [📄 Paper](#) | [🔗 Code](#)
- **Md Imam Ahasan**, Guangchao Yang*, A F M Abdun Noor, S M Hasan Mahmud, Md Mahfuzur Rahman. *Dose-Aware Cold Diffusion with Physics Consistency for Generalizable Low-Dose CT Reconstruction* - International Joint Conference on Neural Networks (IJCNN 2026) - **CCF-C**. [📄 Paper](#) | [🔗 Code](#)
- Ahtisham Waheed, Yunfei Yin*, A F M Abdun Noor, **Md Imam Ahasan**. *Direction-Curvature Aware Feature Integration for Robust Lane Detection* - Computers, Materials and Continua (CMC) - **SCIE**. [📄 Paper](#) | [🔗 Code](#)
- Md Mahfuzur Rahman, Pengzhan Zhou*, A F M Abdun Noor, **Md Imam Ahasan**. *MTF-Net: Multi-Modal Temporal Feature Fusion Network for Pedestrian Intention Prediction* - International Conference on Cloud and Network Computing (ICNC 2026) - **EI Index**. [📄 Paper](#) | [🔗 Code](#)
- Anum Malik, Mahnoor Buriro, Muhammad Younas Khan, **Md Imam Ahasan**. *MoSSGate: Memory-Modulated State-Space Gating for Skin Lesion Segmentation* - International Conference on Cloud and Network Computing (ICNC 2026) - **EI Index**. [📄 Paper](#) | [🔗 Code](#)

Under Review

- **Md Imam Ahasan**, Guangchao Yang*, A F M Abdun Noor, S M Hasan Mahmud*. *GenDiff: A Dose and Anatomy Aware Diffusion Model with Structural Prior Refinement for Low-Dose CT Reconstruction and Generalization* - PeerJ Computer Science - **SCIE**. [📄 Paper](#) | [🔗 Code](#)
- **Md Imam Ahasan**, Guangchao Yang*, A F M Abdun Noor, Mohammad Azam Khan*. *LightGAN-LD: A Lightweight Generative Adversarial Network for Efficient Low-Dose CT Reconstruction with Sinogram Encoding and Edge-Aware Learning* - PeerJ Computer Science - **SCIE**. [📄 Paper](#) | [🔗 Code](#)
- A F M Abdun Noor, **Md Imam Ahasan**, Kah Ong Michael Goh*, S M Hasan Mahmud. *QBiLoF-U-Net: Bi-Context Location-Aware Feature Modeling for Joint Mammographic Segmentation and Classification* - Computer Modeling

in Engineering and Sciences (CMES) - **SCIE**.

 [Paper](#) |  [Code](#)

- Md Mustafizur Rahman, Guangchao Yang, A F M Abdun Noor, **Md Imam Ahasan**, Md Mahfuzur Rahman. *TSCA-Net: Temporal-Spatial Clique Attention for Interpretable Multimodal Pedestrian Trajectory Prediction* - IEEE International Conference on Data Mining (ICDM 2026) - **CCF-B**.  [Paper](#) |  [Code](#)
- **Md Imam Ahasan**, A F M Abdun Noor, Kah Ong Michael Goh*, S M Hasan Mahmud*. *Toward Real-Time LDCT Reconstruction: A Sinogram-Encoded Lightweight GAN with Measurement-Domain Noise Suppression* - Computers, Materials and Continua (CMC) - **SCIE**.  [Paper](#) |  [Code](#)
- **Md Imam Ahasan**, Guangchao Yang*, A F M Abdun Noor, Mohammad Azam Khan. *PCDiff: Physics-Conditioned Bayesian Diffusion Model for Low-Dose CT Reconstruction* - International Conference on Distributed Artificial Intelligence (DAI 2026) - **CCF-B**.  [Paper](#) |  [Code](#)
- Md Mahfuzur Rahman, Pengzhan Zhou*, A F M Abdun Noor, **Md Imam Ahasan**, Md Mustafizur Rahman. *Adaptive Cross-Modal Fusion with Sparse Attention for Pedestrian Crossing Intention Prediction* - PeerJ Computer Science - **SCIE**.  [Paper](#) |  [Code](#)

Research Experience

Graduate Research Assistant, Computer Vision Lab

Chongqing, China

Chongqing University

Sep 2024 – Present

Supervisor: Dr. Guangchao Yang, College of Computer Science

Designed **DACD**, a Poisson-thinning cold diffusion model with physics consistency achieving +1.20 dB PSNR over SOTA and 3.3× faster inference. Extended to **GenDiff** with dose-anatomy encoding and structural prior refinement for zero-shot generalization across unseen dose levels and anatomical domains.

Lab Assistant, Data Science Lab

Dhaka, Bangladesh

Daffodil International University

Jun 2020 – Jul 2021

Supervisor: Dr. Md Zahid Hasan, Computer Science and Technology

Built and evaluated machine learning and deep learning models for tasks such as breast cancer classification and other Kaggle-based datasets, covering data preprocessing, model training, and performance evaluation.

Technical Skills

Languages	Python
Frameworks	PyTorch, MONAI, scikit-learn
Computer Vision	Classification, segmentation, generative models, inverse problems
Tools	Git, Linux, LaTeX, experiment tracking (W&B)

Teaching & Professional Experience

Senior IT Support Engineer

Dhaka, Bangladesh

Surma Garments Ltd.

Aug 2022 – Jun 2023

Lecturer, (ICT)

Dhaka, Bangladesh

Anupama International School and College

Dec 2021 – Jun 2022

Awards, Leadership & Community

CQU Outstanding International Graduate, Chongqing University

June 2026

3rd Prize, “China in My Eyes” Contest

Oct 2024

Finalist, Joy Bangla Youth Award

Aug 2021

Best Club Performance Award, Daffodil International University

Feb 2020

Co-Founder, Shopner Khoje Foundation (Charity Organization)

Feb 2017 – Present

Service & Volunteering

Peer Reviewer for: Engineering Applications of Artificial Intelligence, Systems & Soft Computing, Journal of Agriculture and Food Research, PRCV2025, PRCV2026, IJCNN2026.

Volunteer: ICCNC 2026, The 2018 ICPC Asia Dhaka Regional Contest, 6th International Conference on Electronic Engineering and Informatics, IEEE International Conference on Electrical Energy Conversion Systems and Control.

References

Dr. Guangchao Yang

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Chongqing University, Chongqing, China.
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Phone: +86 15923504788

Dr. Mohammad Azam Khan

Kim Jaechul Graduate School of AI,
Korea Advanced Institute of Science and Technology.
Email: azamkhan@kaist.ac.kr